

QUADRUPEd ROBOT PROTOTYPE

TECHNICAL DESIGN AND ASSEMBLY REPORT

Project	Quadruped Robot Prototype
Organisation	BHIV Robotics
Document Type	Technical Design and Assembly Report
SolidWorks File	quadruped robot dog.SLDASM
Phase	Phase 1 Design Freeze (Sprint Fabrication Phase)

1. EXECUTIVE SUMMARY

This report documents the complete technical design, assembly structure, mass analysis, material selection, and dimensional details of the ARYA 60% scale quadruped robot prototype developed by BHIV Robotics. The prototype is a four-legged walking robot designed to achieve stable locomotion using a crawl gait, with each leg driven by two MG996R servo motors providing hip and knee pitch degrees of freedom.

The structural components are fabricated from Nylon 101 using FDM 3D printing. The total structural mass confirmed from SolidWorks Mass Properties is 798.37 g. The full assembly has been completed in SolidWorks with all mates resolved, mass properties verified, and manufacturing drawings produced. This document serves as the Phase 1 Design Freeze technical record.

1.1 Key Design Parameters

Parameter	Value
Overall Length	262.77 mm
Overall Width	213.22 mm
Overall Height	186.23 mm
Total Structural Mass (CAD)	798.37 g (confirmed from SolidWorks)
Estimated All-Up Mass (with electronics)	Approx 1.2 kg
Total Weight Force	7.834 N (0.79837 kg x 9.81 m/s ²)

Centre of Mass X	58.06 mm from origin
Centre of Mass Y	50.61 mm from origin
Centre of Mass Z	79.11 mm from origin
Number of Legs	4
Degrees of Freedom	8 (2 per leg: hip pitch + knee pitch)
Gait Type	Crawl gait (3-leg support model)
Structural Material	Nylon 101 PA12 (FDM 3D printed)
Foot Pad Material	Rubber / TPU 95A
Servo Type	8 x MG996R Metal Gear
Controller	Raspberry Pi 4 Model B 4 GB
Servo Driver	PCA9685 16-channel PWM via I2C

Battery	3S LiPo 11.1 V 2200 to 3000 mAh
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2. SYSTEM OVERVIEW AND ARCHITECTURE

2.1 Robot Architecture

The ARYA quadruped robot follows a ladder-frame chassis architecture with an open electronics bay. The chassis consists of an upper plate and lower plate connected by vertical posts and cross-braces, forming a rigid box frame. Four legs attach at the four corners of the chassis through hip servo brackets. Each leg consists of a thigh link driven by the hip servo and a shin link driven by the knee servo, connected by a knee bracket. The foot connector and rubber foot pad form the ground-contact element at the bottom of each leg.

2.2 Assembly Hierarchy

Level	Subassembly	Components Included	Material	Qty
Level 0	Full Assembly	quadruped robot dog.SLDASM	All subassemblies	1
Level 1	Chassis Subassembly	Upper plate + Lower plate + Posts + Electronics plate	Nylon 101	1 set
Level 1	Front-Left Leg	Hip bracket + Thigh + Knee bracket + Shin + Foot	Nylon 101 + TPU	1
Level 1	Front-Right Leg	Hip bracket + Thigh + Knee bracket + Shin + Foot	Nylon 101 + TPU	1
Level 1	Rear-Left Leg	Hip bracket + Thigh + Knee bracket + Shin + Foot	Nylon 101 + TPU	1

Level 1	Rear-Right Leg	Hip bracket + Thigh + Knee bracket + Shin + Foot	Nylon 101 + TPU	1
Level 2	Electronics Subassembly	Raspberry Pi + PCA9685 + IMU + Camera + BMS	PCB modules	1 set
Level 2	Power Subassembly	Battery + BMS + Buck converter + Distribution board	Mixed	1 set

2.3 Load Transfer Path

Ground Contact Force -> Rubber Foot Pad -> Foot Connector -> Shin Link -> Knee Servo Bracket -> Thigh Link -> Hip Servo Bracket -> Chassis Lower Plate -> Chassis Upper Plate

This path ensures that structural loads are carried by the printed Nylon 101 frame rather than passing directly through the servo motor housings.

3. INDIVIDUAL PARTS - DIMENSIONS AND MATERIAL

3.1 Chassis Components

Sr.	Part Name	Material	Dimensions (mm)	Qty	Print Settings	Function	Est. Mass
13	Upper Chassis Plate	Nylon 101	262.77 x 213.22 x 6	1	40% gyroid, 4 walls, 0.20 mm layer	Top structural plate, mounting surface for electronics and posts	~85 g
14	Lower Chassis Plate	Nylon 101	262.77 x 213.22 x 6	1	40% gyroid, 4 walls, 0.20 mm layer	Bottom structural plate, mounts to hip brackets at 4 corners	~75 g

15	Electronics Plate	Nylon 101	180 x 100 x 4	1	30% infill, 3 walls, 0.20 mm layer	Internal plate for Raspberry Pi, PCA9685, BMS mounting	~35 g
22	Battery Tray	Nylon 101	115 x 45 x 30	1	40% infill, 4 walls, 0.20 mm layer	Holds 3S LiPo battery securely inside chassis bay	~20 g
23	Camera Mount	Nylon 101	40 x 35 x 25	1	40% infill, 3 walls, 0.20 mm layer	Front-face bracket for CSI camera module	~10 g
24	IMU Holder	Nylon 101	25 x 25 x 10	1	50% infill, 4 walls, 0.20 mm layer	Rigid mount for IMU at centre of mass location	~5 g

3.2 Hip Joint Components (x4 per robot)

Sr.	Part Name	Material	Dimensions (mm)	Qty	Type	Function	Est. Mass
16	Hip Servo Bracket	Nylon 101	40 x 35 x 25 (outer), 3 mm wall	4	50% infill, 4 walls, 0.15 mm layer	Houses MG996R hip servo, mounts to chassis corner via 4x M3 screws	~22 g each
10	MG996R Hip Servo	Metal + Plastic case	40 x 20 x 43	4	Purchased component	Drives hip pitch rotation, 11 kg.cm at 6 V	~55 g each

12	Servo Horn	Aluminium or Plastic	Diameter 24 mm	4	Purchased component	Output arm from hip servo, connects to thigh link	~5 g each
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3.3 Thigh Link (x8 per robot - 2 parallel per leg)

Sr.	Part Name	Material	Dimensions (mm)	Qty	Print Settings	Function and Notes	Est. Mass
18	Thigh Link	Nylon 101	80 x 15 x 8	8	40% infill, 4 walls, 0.20 mm layer	Connects hip servo horn to knee bracket. Axis-to-axis length 80 mm. Two parallel links per leg for lateral rigidity. 3.2 mm bore at each end for M3 clevis pin.	~12 g each

3.4 Knee Joint Components (x4 per robot)

Sr.	Part Name	Material	Dimensions (mm)	Qty	Type	Function	Est. Mass
17	Knee Servo Bracket	Nylon 101	35 x 30 x 25 (outer), 3 mm wall	4	50% infill, 4 walls, 0.15 mm layer	Houses MG996R knee servo, pivots on thigh link via clevis pin	~18 g each
11	MG996R Knee Servo	Metal + Plastic case	40 x 20 x 43	4	Purchased component	Drives knee pitch rotation, 11 kg.cm at 6 V	~55 g each

12	Servo Horn	Aluminium or Plastic	Diameter 24 mm	4	Purchased component	Output arm from knee servo, connects to shin link	~5 g each
27	Clevis Pin ISO 2341	Mild steel	Diameter 6 x 30 mm	8	Purchased component	Pivot pin connecting thigh to knee bracket	~3 g each

3.5 Shin Link (x8 per robot - 2 parallel per leg)

Sr.	Part Name	Material	Dimensions (mm)	Qty	Print Settings	Function and Notes	Est. Mass
19	Shin Link	Nylon 101	70 x 15 x 8	8	40% infill, 4 walls, 0.20 mm layer	Connects knee servo horn to foot connector. Axis-to-axis length 70 mm. Two parallel links per leg. 3 mm internal bore runs full length for wiring channel. 3.2 mm end bores for clevis pins.	~10 g each

3.6 Foot Assembly (x4 per robot)

Sr.	Part Name	Material	Dimensions (mm)	Qty	Type	Function	Est. Mass
20	Foot Connector	Nylon 101	20 x 12 x 12	4	50% infill, 4 walls, 0.15 mm layer	Structural link between shin and foot pad. Has 13 mm dia sensor pocket for FSR foot contact sensor. Pin bore at top for clevis pin.	~8 g each

21	Rubber Foot Pad	TPU 95A	Diameter 25 x 6 mm hemispherical	4	30% infill, 3 walls, 0.20 mm layer	Ground contact element. Hemispherical profile enables passive terrain adaptation. Press-fit onto foot connector.	~6 g each

3.7 Fasteners and Hardware Summary

Sr.	Component	Standard	Size	Qty	Type	Purpose
25	Socket Head Cap Screw M3 x 8 mm	ISO 4762 Stainless A2	M3 x 8 mm	50	Purchased	Servo mounting to brackets, cover plates
26	Socket Head Cap Screw M3 x 12 mm	ISO 4762 Stainless A2	M3 x 12 mm	20	Purchased	Link to servo horn connections
27	Socket Head Cap Screw M3 x 16 mm	ISO 4762 Stainless A2	M3 x 16 mm	20	Purchased	Chassis to bracket mounting
28	Hex Nut M3	ISO 4032 Stainless	M3	52	Purchased	Counter-nut for all screw positions
29	Flat Washer M3	ISO 7089 Stainless	M3	52	Purchased	Load distribution under screw heads
30	Spring Washer M3	DIN 127 Stainless	M3	52	Purchased	Vibration resistance at all joints

31	Brass Heat-Set Insert M3	M3 x 5 mm x 4 mm OD	M3 x 5 mm	24	Purchased	Threaded inserts in all Nylon 101 printed parts
32	Clevis Pin ISO 2341	Mild steel	Diameter 6 x 30 mm	8	Purchased	Pivot pins at knee and foot joints
43	Ball Bearing MR106ZZ	Steel shielded	6 x 10 x 3 mm	8	Purchased (optional)	Idler bearings at joint pivot pins

4. MASS AND WEIGHT CALCULATIONS

4.1 SolidWorks Confirmed Mass Properties

The following values are directly extracted from the SolidWorks Mass Properties dialog for the complete assembly file "quadruped robot dog.SLDASM". These are the authoritative values for all downstream calculations.

Property	Value (from SolidWorks)
Total Assembly Mass	798.37 g
Volume	695,562.74 cubic millimetres
Surface Area	385,491.20 square millimetres

Centre of Mass - X coordinate	58.06 mm
Centre of Mass - Y coordinate	50.61 mm
Centre of Mass - Z coordinate	79.11 mm
Principal Moment of Inertia Px	3,271,924.82 g.mm ²
Principal Moment of Inertia Py	6,964,035.46 g.mm ²
Principal Moment of Inertia Pz	7,761,359.61 g.mm ²
Ixx at CoM	3,277,281.04 g.mm ²
Iyy at CoM	7,755,214.17 g.mm ²
Izz at CoM	6,964,824.68 g.mm ²
Ixx at output origin	10,319,659.50 g.mm ²
Iyy at output origin	15,443,441.98 g.mm ²

lzz at output origin	11,701,242.0 g.mm2
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4.2 Component Mass Breakdown

Category	Mass	Source	Notes
Nylon 101 Structural Parts (CAD confirmed)	798.37 g	SolidWorks Mass Properties	This is the confirmed CAD value for all printed and assembled structural parts only
MG996R Servo Motors x 8	8 x 55 g = 440 g	Manufacturer datasheet	Each MG996R servo weighs approximately 55 g
Raspberry Pi 4 Model B 4 GB	46 g	Manufacturer datasheet	RPi 4B board only, without case
PCA9685 Servo Driver + Buck Converter + DC Switch	Approx 35 g	Estimate	Small PCB modules combined
3S LiPo Battery 2200 mAh	Approx 180 g	Typical 3S 2200 mAh pack	Varies by brand; measure actual battery before finalising
Wiring, Connectors, Cable Management	Approx 50 g	Estimate	Based on wire lengths and connector counts from BOM

Fasteners (screws, nuts, washers, inserts, pins)	Approx 40 g	Estimate	Based on fastener count and standard M3 hardware masses
TOTAL ALL-UP MASS (estimated)	Approx 1189 g = approx 1.2 kg	Design target: under 1.5 kg	Actual must be physically weighed after Phase 6 full assembly

4.3 Structural Load Calculations

All calculations use the SolidWorks confirmed structural mass of 798.37 g as the basis. The estimated all-up mass of 1.2 kg is used for worst-case design load scenarios.

Load Case	Calculation and Result	Notes
Basic Weight (structural mass only)	$W = m \times g = 0.79837 \times 9.81 = 7.834 \text{ N}$	—
Basic Weight (all-up mass, 1.2 kg)	$W = 1.2 \times 9.81 = 11.772 \text{ N}$	Used for worst-case design
Static load per leg (4-leg stand, all-up)	$F = 11.772 / 4 = 2.943 \text{ N per leg}$	All 4 legs on ground
Walking load per leg (crawl gait, 3-leg support)	$F = 11.772 / 3 = 3.924 \text{ N per leg}$	One leg lifted, 3 support

Worst case (1 leg, 50% of all-up weight, uneven terrain)	$F = 11.772 \times 0.50 = 5.886 \text{ N}$	Extreme terrain scenario
Dynamic load factor applied (walking)	Design load = $3.924 \times 2.0 = 7.848 \text{ N}$	Dynamic factor = 2.0
Impact load factor applied (slip or landing)	Impact load = $5.886 \times 2.5 = 14.715 \text{ N}$	Impact factor = 2.5

4.4 Joint Torque Calculations

Link lengths used: Thigh link axis-to-axis = 80 mm, Shin link axis-to-axis = 70 mm. These are placeholder values pending Many's kinematic sign-off.

Case	Calculation and Result	Notes
Hip Torque (crawl gait walking)	$T = F_{\text{leg}} \times L_{\text{thigh}} = 3.924 \times 0.080 = 0.314 \text{ N.m}$	—
Hip Torque (dynamic factor x2)	$T = 0.314 \times 2.0 = 0.628 \text{ N.m}$	Design torque at hip
Hip Torque (impact factor x2.5)	$T = 0.314 \times 2.5 = 0.785 \text{ N.m}$	Extreme case at hip

Knee Torque (crawl gait walking)	$T = F_{\text{leg}} \times L_{\text{shin}} = 3.924 \times 0.070 = 0.275 \text{ N.m}$	—
Knee Torque (dynamic factor x2)	$T = 0.275 \times 2.0 = 0.550 \text{ N.m}$	Design torque at knee
Knee Torque (impact factor x2.5)	$T = 0.275 \times 2.5 = 0.688 \text{ N.m}$	Extreme case at knee

4.5 Servo Torque Validation

Joint	Servo	Capacity	Design Torque	Impact Torque	Safety Margin	Status
Hip Joint	MG996R at 6 V	1.08 N.m	0.628 N.m (design)	0.785 N.m (impact)	1.72x margin (design)	SAFE
Knee Joint	MG996R at 6 V	1.08 N.m	0.550 N.m (design)	0.688 N.m (impact)	1.96x margin (design)	SAFE

Safety margins calculated using design load. Under impact loading margins remain above 1.0x. For improved margins use 7.4V battery supply which raises MG996R capacity to approximately 1.37 N.m.

5. MATERIAL SPECIFICATION

5.1 Nylon 101 (PA12) - Primary Structural Material

Nylon 101 was selected as the primary material for all load-bearing printed components due to its superior fatigue resistance, low moisture absorption compared to standard Nylon 6 or 12, and improved dimensional stability post-print versus PLA or PETG. It is the recommended engineering-grade Nylon for functional FDM parts.

Property	Value
Material Family	Polyamide 12 (PA12)
Trade Name	Nylon 101 / Markforged Nylon or equivalent PA12
Density	1.01 to 1.14 g/cm3
Tensile Strength (FDM printed)	Approx 82 MPa (0 degree orientation)
Yield Strength (FDM printed)	Approx 55 MPa
Flexural Modulus	Approx 2,400 MPa
Elongation at Break	20 to 35%
Water Absorption	Low (0.25% in 24 h vs 1.6% for PA6)
Heat Deflection Temperature	Approx 80 degrees C (0.45 MPa load)
Factor of Safety Applied	2.0 (for prototype phase)

Allowable Design Stress	55 / 2.0 = 27.5 MPa
Nozzle Temperature	250 to 260 degrees C
Bed Temperature	70 to 80 degrees C
Enclosure Required	Yes (prevents warping)
Drying Required	80 degrees C for 6 to 8 hours before printing
Recommended Infill (structural)	40 to 50% gyroid
Recommended Wall Count	4 perimeters minimum for joints
Recommended Layer Height	0.15 mm for joints, 0.20 mm for links and plates

5.2 Structural Stress Check - Thigh Link

The thigh link is the most critically loaded structural member. The following check confirms it is safe under design load.

Parameter	Value	Notes	Status
Link material	Nylon 101 PA12	—	—

Yield strength	55 MPa	—	—
Allowable stress (FoS = 2.0)	27.5 MPa	—	—
Cross-section	C-channel, 15 mm wide x 8 mm tall, 2.5 mm wall	Net area approx 55 mm ²	—
Design axial load	7.848 N (dynamic, all-up mass basis)	—	—
Calculated stress $\sigma = F / A$	$7.848 / 55 = 0.143$ MPa	—	0.143 MPa << 27.5 MPa - SAFE
Stress safety factor	$27.5 / 0.143 = 192x$	Very high margin	Thigh link is not the failure mode

5.3 Buckling Check - Shin Link

Parameter	Value	Notes	Status
Euler buckling load formula	$P_{cr} = (\pi^2 \times E \times I) / (K \times L)^2$	K = 1.0 (pin-pin)	—

Modulus E	2,400 MPa = 2,400 N/mm ²	Nylon 101 flexural modulus	—
Second moment of area I	For Dia 6 mm circular: $I = \pi \times 6^4 / 64 = 63.6 \text{ mm}^4$	—	—
Effective length K x L	1.0 x 70 = 70 mm	—	—
Critical buckling load P_{cr}	$(9.87 \times 2400 \times 63.6) / (70)^2 = 308 \text{ N}$	—	—
Design load on shin link	14.715 N (impact case, all-up mass)	—	—
Buckling safety factor	$308 / 14.715 = 20.9x$	—	SAFE - No buckling risk

5.4 TPU 95A - Foot Pad Material

Property	Value
Material	Thermoplastic Polyurethane Shore 95A
Purpose	Ground contact, shock absorption, grip

Shore Hardness	95A (firm but compressible)
Tensile Strength	Approx 40 to 50 MPa
Elongation at Break	400 to 600%
Print Temperature	220 to 240 degrees C
Infill	30% (softer response at lower infill)
Geometry	Hemispherical tip diameter 25 mm x 6 mm height

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6.3 Fastener Torque Specifications

Location	Torque Specification	Critical Note
M3 x 8 into Nylon 101 heat-set insert	0.5 N.m maximum	Overtorque strips the brass insert from the Nylon 101 bore

M3 x 12 into Nylon 101 heat-set insert	0.5 N.m maximum	Check insert is fully seated before tightening
M3 x 16 into Nylon 101 heat-set insert	0.5 N.m maximum	Chassis bolts: use spring washer at all positions
Clevis pin retaining clip	Hand press only	Do not use hammer - deforms clip groove
Bearing press-fit into housing	Manual press with flat rod only	Never hammer directly on bearing race
Servo horn screw (M2.5 self-tapping)	0.3 N.m maximum	Stripped horn = loss of servo output - check before every test

7. CENTRE OF MASS ANALYSIS AND STABILITY

7.1 CoM Position

The SolidWorks confirmed centre of mass for the structural assembly is at X = 58.06 mm, Y = 50.61 mm, Z = 79.11 mm from the assembly origin.

Axis	CoM Position (mm)	Geometric Centre (mm)	Offset (mm)
X (length direction)	58.06	$262.77 / 2 = 131.39$	73.33 mm forward of geometric centre

Y (width direction)	50.61	$213.22 / 2 = 106.61$	56.00 mm to one side of geometric centre
Z (height direction)	79.11	$186.23 / 2 = 93.12$	14.01 mm below geometric centre (good)

7.2 CoM Offset Assessment

The X-axis offset of 73.33 mm and Y-axis offset of 56.00 mm indicate the CoM is shifted significantly forward and to one side of the body geometric centre. This is expected given the electronics and battery placement. The critical check is whether the CoM falls within the support polygon formed by the 4 foot contact points in the standing pose.

ACTION REQUIRED: Before Phase 1 sign-off, the standing pose must be simulated in SolidWorks with legs in position. The CoM must be projected down onto the ground plane and verified to fall within the quadrilateral formed by the 4 foot contact points. If it falls outside, the battery tray must be shifted rearward by approximately 30 to 40 mm to rebalance.

8. 3D PRINTING SPECIFICATIONS

8.1 Recommended Print Settings by Part Category

Part Category	Material	Layer Height	Infill	Walls	Nozzle Temp	Bed Temp	Enclosure	Est. Time
Chassis Plates (upper, lower)	Nylon 101	0.20 mm	40% gyroid	4	250 to 260 C	70 to 80 C	Required	4 to 5 hrs each
Hip and Knee Brackets	Nylon 101	0.15 mm	50% gyroid	4	250 to 260 C	70 to 80 C	Required	1.5 to 2 hrs each
Thigh and Shin Links	Nylon 101	0.20 mm	40% gyroid	4	250 to 260 C	70 to 80 C	Required	45 min to 1 hr each

Foot Connector	Nylon 101	0.15 mm	50% gyroid	4	250 to 260 C	70 to 80 C	Required	30 min each
Foot Pad	TPU 95A	0.20 mm	30% gyroid	3	220 to 240 C	30 to 40 C	Not required	20 min each
Electronics and Battery Plates	Nylon 101	0.20 mm	30% rectilinear	3	250 to 260 C	70 to 80 C	Required	1 to 2 hrs each
IMU Holder, Camera Mount	Nylon 101	0.20 mm	50% gyroid	4	250 to 260 C	70 to 80 C	Required	20 to 45 min each

Critical: Dry Nylon 101 at 80 degrees C for minimum 6 to 8 hours before printing. Moisture causes bubbling, poor layer bonding, and dimensional inaccuracy. Use sealed filament bags with desiccant for storage

END OF REPORT - Quadruped Robot Prototype - Technical Design and Assembly Report - Rev 1.0 - July 2026